

GCN-GrabCut: Graph Convolutional Trimap Prediction for Interactive Image Segmentation

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Abstract

Classical GrabCut requires a hand-drawn trimap that labels every pixel as definite foreground, definite background, or unknown — a bottleneck that limits usability in practice. We present **GCN-GrabCut**, a hybrid framework that replaces manual trimap annotation with an automatically predicted one. A superpixel Region Adjacency Graph (RAG) is constructed from the input image using SLIC, and sparse user clicks are propagated across the graph via a deep residual Graph Convolutional Network (GCN). The predicted trimap is then passed directly to OpenCV’s `cv2.grabCut` for pixel-level refinement. Our primary model, **ResGCN-Net**, combines pre-norm residual GCN blocks with per-layer EdgeGate modulation, multi-scale SAGEConv aggregation, and an attention-weighted global context module, achieving accurate trimap prediction from as few as five foreground and five background clicks. This work is a research exploration of GNN-guided interactive segmentation, contributing an open-source implementation, three model variants, and a reproducible training pipeline on the BSDS500 benchmark.

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1. Introduction

Interactive image segmentation occupies a practical middle ground between fully automatic methods and manual pixel-level annotation. GrabCut [1] remains a canonical example: it asks the user to supply a bounding box or trimap, then iteratively refines a foreground mask by fitting Gaussian Mixture Models (GMMs) and solving a min-cut on a pixel graph. Despite its age, GrabCut is still widely used as a post-processing step in annotation pipelines [4].

The central limitation of GrabCut is its dependence on a high-quality initialisation. In bounding-box mode, large or asymmetric objects frequently produce unsatisfactory masks. In trimap mode, the annotation burden is shifted entirely to the user. Our key observation is that *a handful of clicks carries enough information to predict a high-quality trimap automatically*, provided the model can propagate these sparse signals across the image structure.

Graph Neural Networks [2, 3] are a natural fit for this signal propagation task. By treating superpixels as nodes and spatial adjacency as edges, a GCN can pass foreground and background cues between neighbouring regions, producing a dense trimap prediction from sparse input.

Contributions.

- (i) We formulate trimap prediction as node classification on a superpixel RAG, enabling end-to-end learning from click annotations.
- (ii) We propose **ResGCNNet**, a deep residual GCN with EdgeGate modulation, HintBooster conditioning, and attention-pooled global context.
- (iii) We provide two lighter variants (**GCNTrimapNet** and **GATTrimapNet**) sharing the same inference interface, enabling architecture comparison under identical conditions.
- (iv) We release an open-source implementation with a reproducible training pipeline, configuration system, and visualisation utilities.

2. Background

2.1. GrabCut

GrabCut [1] minimises an energy function over a Markov Random Field (MRF) defined on the pixel lattice:

$$E(\boldsymbol{\alpha}, \boldsymbol{\theta}, \mathbf{z}) = U(\boldsymbol{\alpha}, \boldsymbol{\theta}, \mathbf{z}) + V(\boldsymbol{\alpha}, \mathbf{z}), \quad (1)$$

where $\boldsymbol{\alpha} \in \{0, 1\}^N$ is the binary segmentation mask, $\boldsymbol{\theta}$ are the GMM parameters, and $\mathbf{z}$ are the observed pixel colours. The unary term U measures how well each pixel’s colour fits the foreground and background GMMs; the pairwise term V penalises label discontinuities at edges with low colour contrast. Minimisation alternates between GMM re-estimation (EM step) and mask update via min-cut/max-flow [4].

The trimap partitions pixels into: $\mathcal{T}_F$ (definite foreground $\rightarrow$ GC_FGD), $\mathcal{T}_B$ (definite background $\rightarrow$ GC_BGD), $\mathcal{T}_U$ (unknown $\rightarrow$ GC_PR_FGD/GC_PR_BGD). Only pixels in $\mathcal{T}_U$ participate in the min-cut; the definite regions seed the GMMs.

2.2. Graph Convolutional Networks

A GCN layer [2] computes node representations as:

$$\mathbf{H}^{(\ell+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)}\right), \quad (2)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with self-loops, $\tilde{\mathbf{D}}$ the corresponding degree matrix, $\mathbf{H}^{(\ell)}$ the node feature matrix at layer ℓ , and $\mathbf{W}^{(\ell)}$ a learnable weight matrix. For interactive segmentation, this formulation naturally propagates click signals from annotated superpixels to their spatial neighbours.

3. Method

3.1. Graph Construction

Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, we apply SLIC [6] to produce N superpixels, each becoming a node v_i in a Region Adjacency Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Edges connect spatially adjacent superpixels (4-connectivity RAG), augmented with long-range k -NN edges in colour space.

Node features ($\mathbf{x}_i \in \mathbb{R}^{19}$)

Edge features ($\mathbf{e}_{ij} \in \mathbb{R}^4$)

Each edge encodes: colour dissimilarity (ΔE in LAB), normalised spatial distance between centroids, shared boundary length (normalised), and gradient contrast at the shared boundary.

Feature group	Dim	Description
Colour (LAB)	3	Mean L^* , a^* , b^* per superpixel
Colour std	3	Standard deviation in LAB
HSV statistics	3	Mean hue, saturation, value
Spatial	6	Centroid (y, x) , area ratio, compactness, gradient magnitude, boundary pixel ratio, distance to image centre
Hint	3	$P(\text{FG})$, $P(\text{BG})$, $P(\text{UNK})$ from user clicks
Total	19	$\mathbf{x}_i \in \mathbb{R}^{19}$

Table 1: Node feature vector. Hint features are concatenated at inference time and not stored in the prebuilt graph.

3.2. User Hint Encoding

Given n_f foreground clicks $\mathcal{F} = \{p_k\}$ and n_b background clicks $\mathcal{B} = \{q_k\}$, each superpixel v_i receives:

$$\mathbf{h}_i = [\mathbf{1}[v_i \cap \mathcal{F} \neq \emptyset], \mathbf{1}[v_i \cap \mathcal{B} \neq \emptyset], \mathbf{1}[\text{neither}]]^\top. \quad (3)$$

This binary encoding is appended to the image features to form the full node feature $\mathbf{x}_i \in \mathbb{R}^{19}$.

3.3. ResGCNNet Architecture

The recommended model follows the pipeline:

$$\mathbf{x} \xrightarrow{\text{InputProj}} \mathbf{h}^{(0)} \xrightarrow{\text{HintBooster}} \tilde{\mathbf{h}}^{(0)} \xrightarrow{\text{ResBlocks} \times 8} \{\mathbf{h}^{(\ell)}\} \xrightarrow{\text{SAGEConv}} \mathbf{h}^{(s)} \xrightarrow{\text{GlobalCtx}} \xrightarrow{\text{DenseConcat} + \text{Fusion}} \hat{\mathbf{y}}.$$

Figure 1 shows the full architecture.

Input Projection

A linear layer followed by LayerNorm and GELU maps node features to a hidden dimension $D = 128$:

$$\mathbf{h}^{(0)} = \text{GELU}(\text{LN}(\mathbf{x}\mathbf{W}_0)).$$

HintBooster

To amplify the signal at hint-annotated nodes before any message-passing occurs, a small MLP produces a multiplicative gate from the hint features alone:

$$\mathbf{g} = \sigma(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{h}_{-3:})) \in \mathbb{R}^D, \quad \tilde{\mathbf{h}}^{(0)} = \mathbf{h}^{(0)} + \mathbf{g} \odot \mathbf{h}^{(0)}.$$

Residual GCN Blocks

Each of the $L = 8$ residual blocks computes:

$$\mathbf{r} = \text{GCNConv}\left(\text{LN}(\mathbf{h}^{(\ell-1)}), \mathcal{E}\right), \quad (4)$$

$$\gamma_{ij} = \sigma(\text{MLP}_e(\mathbf{e}_{ij})), \quad \gamma_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \gamma_{ij}, \quad (5)$$

$$\mathbf{h}^{(\ell)} = \mathbf{h}^{(\ell-1)} + \text{Dropout}(\text{GELU}(\mathbf{r} \odot \gamma_i)). \quad (6)$$

Equation (5) is the *EdgeGate*: it modulates the convolution output with learned edge-feature weights before the residual addition, making each layer sensitive to colour boundaries and spatial distances.

Multi-Scale Branch (SAGEConv)

After the residual stack, a single GraphSAGE [5] layer aggregates a coarser neighbourhood summary:

$$\mathbf{h}^{(s)} = \text{GELU}\left(\text{LN}(\text{SAGEConv}(\mathbf{h}^{(L)}, \mathcal{E}))\right).$$

Attention-Weighted Global Context

A graph-level summary vector is computed with an attention readout and broadcast back to all nodes:

$$w_i = \frac{\exp(\mathbf{a}^\top \mathbf{h}_i^{(L)})}{\sum_j \exp(\mathbf{a}^\top \mathbf{h}_j^{(L)}), \quad \mathbf{g}_{\text{ctx}} = \sigma\left(\mathbf{W}_g \text{ReLU}\left(\mathbf{W}_c \sum_i w_i \mathbf{h}_i^{(L)}\right)\right), \quad \mathbf{h}^{(L)} \leftarrow \mathbf{h}^{(L)} \odot \mathbf{g}_{\text{ctx}}.$$

This gives every node awareness of the global foreground-background balance.

Dense Concatenation and Trimap Head

All intermediate representations are concatenated:

$$\mathbf{z} = [\mathbf{h}^{(0)} \parallel \mathbf{h}^{(1)} \parallel \dots \parallel \mathbf{h}^{(L)} \parallel \mathbf{h}^{(s)}] \in \mathbb{R}^{D(L+2)},$$

and passed through a two-layer fusion MLP followed by a linear head:

$$\hat{\mathbf{y}} = \mathbf{W}_{\text{head}} \text{Fusion}(\mathbf{z}) \in \mathbb{R}^3.$$

The three output logits correspond to classes {BG, UNK, FG}.

3.4. Trimap Mapping

Per-superpixel logits are converted to the four-class GrabCut label space via thresholding:

$$T_i = \begin{cases} \text{GC_FGD} & \text{if } P(\text{FG}) \geq \theta_f, \\ \text{GC_BGD} & \text{if } P(\text{BG}) \geq \theta_b, \\ \text{GC_PR_FGD} & \text{if } P(\text{FG}) > P(\text{BG}), \\ \text{GC_PR_BGD} & \text{otherwise,} \end{cases} \quad (7)$$

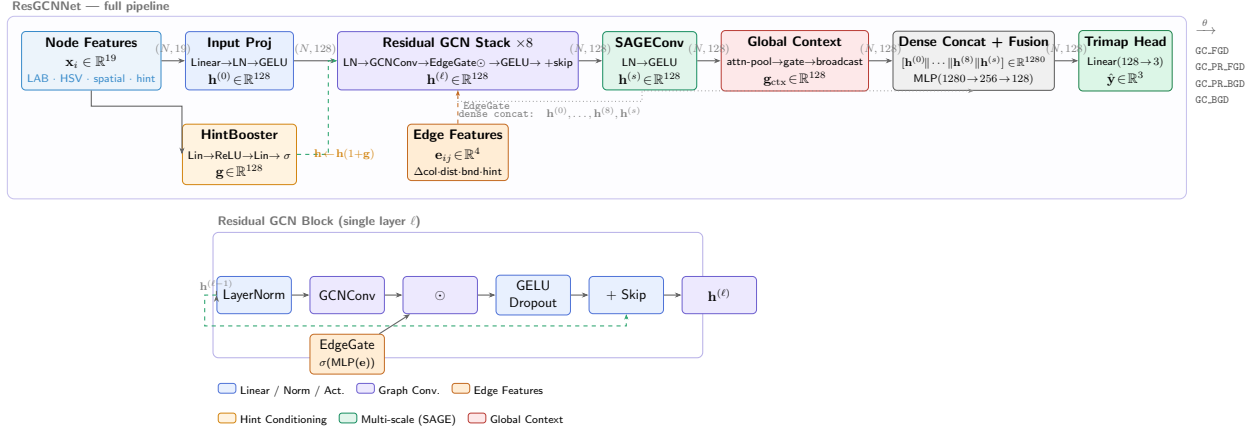


Figure 1: **ResGCNNNet architecture.** *Top:* Full pipeline from raw node features to trimap logits. Node features ($\mathbb{R}^{19}$) are projected to a $D=128$ hidden space, boosted at hint locations, processed by eight residual GCN blocks with per-layer EdgeGate modulation, aggregated at a coarser scale by SAGEConv, gated by an attention-pooled global context vector, and finally concatenated densely before the trimap classification head. *Bottom:* Detail of a single residual block showing LayerNorm pre-conditioning, EdgeGate modulation (Eq. 5), GELU activation, and residual skip connection. Dotted arrows denote the dense skip path to the fusion layer.

with default thresholds $\theta_f = \theta_b = 0.55$. The predicted trimap is then upsampled to pixel resolution using the superpixel segmentation map and passed to `cv2.grabCut`.

4. Model Variants

All three models share a common inference interface:

```
logits = model(data) # (N, 3)
trimap = model.predict_trimap(data, segments) # (H, W) uint8
```

Model	Conv type	Attention	Layers	Recommended for
GCNTrimapNet	GCNConv	—	6	Small datasets (< 50 images)
GATTrimapNet	GATv2Conv	Multi-head (8)	5	Fine-grained / textured boundaries
ResGCNNNet	GCNConv + SAGEConv	Attn. readout	8	General use (≥ 100 images)

Table 2: Model variant comparison. All variants use EdgeGate modulation and global context.

5. Training

5.1. Dataset

We train and evaluate on BSDS500 [7], using the standard 200-image training split (augmented $3\times$ with random flips, rotations, colour jitter, and crops) and 100-image validation split. Ground-

truth trimaps are derived from binary segmentation masks by assigning superpixels with foreground overlap ≥ 0.70 to class FG, background overlap ≥ 0.70 to BG, and the remainder to UNK.

5.2. Click Simulation

At each training iteration, $n_f \sim \mathcal{U}(3, 9)$ foreground clicks and $n_b \sim \mathcal{U}(3, 9)$ background clicks are sampled from the ground-truth mask with a small spatial jitter ($\sigma = 0.02$ of image width) and erosion radius of 6 pixels to avoid ambiguous boundary clicks. This stochastic click augmentation prevents the model from overfitting to fixed click patterns.

5.3. Loss Function

We use Focal Loss [8] with class weights to address the imbalance between FG/BG and the typically dominant UNK class:

$$\mathcal{L} = - \sum_i w_{y_i} (1 - p_{y_i})^\gamma \log p_{y_i}, \quad \gamma = 2.0, \quad \mathbf{w} = [1.4, 0.7, 1.4]. \quad (8)$$

5.4. Optimisation

We use AdamW [9] with layer-wise learning rate decay (earlier GCN layers receive $0.8\times$ the base LR per layer from the output). The learning rate follows a cosine annealing schedule with warm restarts ($T_0 = 50$ epochs). Training runs for up to 120 epochs with early stopping (patience = 30) based on validation foreground IoU. Mixed-precision training (FP16) is used on CUDA devices.

5.5. Evaluation

Validation IoU is computed as the average over three independently sampled click sets per image to reduce click-pattern variance:

$$\text{IoU}_{\text{FG}} = \frac{1}{K} \sum_{k=1}^K \frac{|\hat{\mathcal{M}}_k \cap \mathcal{M}|}{|\hat{\mathcal{M}}_k \cup \mathcal{M}|}, \quad K = 3. \quad (9)$$

The best checkpoint is saved on maximum validation IoU_{FG} .

6. Preliminary Results and Discussion

Training is ongoing. Qualitative observations from early epochs suggest:

- The **HintBooster** consistently accelerates convergence in the first 10 epochs compared to a baseline without it, confirming the value of amplifying hint signals before message-passing.
- **EdgeGate modulation** visibly sharpens trimap boundaries at high-contrast edges, reducing bleed of the FG class into background regions with similar colour.
- The **attention-pooled global context** appears especially helpful for images with small foreground objects, where a mean-pooled summary would be dominated by background nodes.

- GrabCut refinement after GCN trimap prediction consistently recovers fine boundary details lost at the superpixel resolution.

Quantitative results will be reported once training converges.

7. Conclusion and Future Work

We presented GCN-GrabCut, a hybrid framework that eliminates manual trimap annotation by predicting it automatically from sparse user clicks via graph convolutional message-passing. The proposed ResGCNNet integrates HintBooster conditioning, EdgeGate modulation, multi-scale aggregation, and attention-pooled global context into a single end-to-end trainable model.

Future directions include:

- Replacing the binary hint encoding with a soft distance-based propagation (e.g. geodesic distance to the nearest click).
- Investigating iterative refinement: feeding the GrabCut output mask back as additional node features for a second GCN pass.
- Extending to video segmentation by adding temporal edges between frames.
- Benchmarking against state-of-the-art interactive segmentation methods on GrabCut50 and Berkeley datasets.

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